Statistical Machine Learning II

Comprehensive Laboratory Documentation

Benchmarking Metaheuristic Optimization, Evolutionary Algorithms, Regularization Techniques, and Feature Dimensionality Reduction

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Executive Summary / Lab Overview: This lab document provides a comprehensive comparative empirical evaluation of Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), Simulated Annealing (SA), Binary-coded Genetic Algorithms (GA), and Regularized Regression frameworks (OLS, Ridge, Lasso, ElasticNet).

1. Problem Statement

Metaheuristic algorithms, evolutionary search methods, and regularization frameworks constitute essential components of modern statistical learning, complex global optimization, and high-dimensional feature selection. Unlike deterministic gradient-based optimizers, nature-inspired metaheuristics offer robust mechanisms for escaping local minima in non-convex, multimodal, or combinatorial search spaces. Simultaneously, constrained regression methods enforce structural sparsity and mitigate multicollinearity in feature-rich modeling domains.

The objective of this laboratory exercise is to conduct a comparative empirical evaluation and algorithmic documentation across five key computational optimization and learning tasks:

- **1. Swarm Intelligence & Function Optimization:** Particle Swarm Optimization (PSO) applied to the dynamic 1D quadratic function and the high-dimensional non-convex Rosenbrock benchmark function.
- **2. Nature-Inspired Combinatorial Optimization:** Ant Colony Optimization (ACO) utilizing artificial pheromone deposition and evaporation for shortest path finding in Traveling Salesman Problems (TSP) graph networks.
- **3. Stochastic Local Search:** Simulated Annealing (SA) incorporating Metropolis-Hastings acceptance probability criteria under controlled geometric cooling schedules.
- **4. Genetic & Evolutionary Computing:** Binary-coded Genetic Algorithms (GA) incorporating roulette wheel selection and single-point crossover operations for continuous parameter bound optimization.
- **5. Regularized Linear Modeling:** Structural comparison of Unregularized Ordinary Least Squares (OLS), Ridge (L2), Lasso (L1), and ElasticNet (L1 + L2) regularization on high-dimensional noisy datasets.

Each formulation is benchmarked using trajectory evolution, convergence rate, execution efficiency, coefficient shrinkage metrics, and objective function optimization values.

2. Description of the Metaheuristics & Algorithms

2.1 Optimization Paradigms & Mathematical Formulations

Algorithm / Technique	Operational Paradigm	Target Application / Function	Primary Parameters & Equations
Particle Swarm Optimization (PSO)	Swarm Intelligence	Continuous Optimization ($f(x) = -x^2 + 5x + 20$ & Rosenbrock)	$Vi^{(t+1)} = w\, Vi^{(t)} + c1\, u1\, (Pbest,i - Xi) + c2\, u2\, (Gbest - Xi)$ $Xi^{(t+1)} = Xi^{(t)} + Vi^{(t+1)}$
Ant Colony Optimization (ACO)	Graph-based Stigmergy	Combinatorial Shortest Path / TSP Graph Routing	$Pij = (\tau ij^{\alpha} \cdot \eta ij^{\beta}) / \sum (\tau ik^{\alpha} \cdot \eta ik^{\beta})$ $\tau ij \leftarrow (1-\psi)\tau ij + \sum \Delta \tau ij^{k}$
Simulated Annealing (SA)	Thermodynamic Local Search	Non-Convex Minimization ($f(x) = x^2 - 6x - 13$)	$P(accept) = \exp(-\Delta f / T)$ if $\Delta f > 0$ else 1.0 $T_{(t+1)} = \alpha \cdot T_t$
Genetic Algorithm (GA)	Evolutionary Parameter Search	Bound-Constrained Optimization ($f(x) = -x^2 + 2x$)	Roulette Selection: $Pi = fi / \sum fk$ Single-Point Crossover & Binary Decoding
L1 / L2 / Elastic Net Regularization	Constrained Regression Optimization	Sparse & Shrinkage Feature Selection on High-Dim Data	$\min_{\beta} (1/2N) \|\|y - X\beta\|\|_2^{\,2} + \alpha\, \rho\, \|\|\beta\|\|_1 + (\alpha(1-\rho)/2)\, \|\|\beta\|\|_2^{\,2}$

2.2 Algorithm Implementations & Code Base Overview

The framework incorporates standardized Python modules (NumPy, Scikit-Learn) for exact stochastic simulation, pheromone graph updating, acceptance probability calculations, and penalized likelihood optimization.

3. Whole Python Implementation Code

The unified execution script runs all metaheuristic, evolutionary, and regularized regression benchmarks, outputting quantitative convergence steps and performance metrics.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge, Lasso, ElasticNet
from sklearn.metrics import mean_squared_error, r2_score

# ========================================
# 1. PARTICLE SWARM OPTIMIZATION (PSO)
# ========================================
def objective_pso(x):
    return -x**2 + 5 * x + 20

def run_pso(particles, c1=1, c2=1, u1=0.213, u2=0.876, w=0, iterations=3):
    x = np.array(particles, dtype=float)
    v = np.zeros(len(particles))
    fitness = objective_pso(x)
    p_best = x.copy()
    p_best_fitness = fitness.copy()
    g_best_idx = np.argmax(p_best_fitness)
    g_best = p_best[g_best_idx]

    for t in range(1, iterations + 1):
        v = w * v + c1 * u1 * (p_best - x) + c2 * u2 * (g_best - x)
        x = x + v
        fitness = objective_pso(x)
        improved = fitness > p_best_fitness
        p_best[improved] = x[improved]
        p_best_fitness[improved] = fitness[improved]
        g_best_idx = np.argmax(p_best_fitness)
        g_best = p_best[g_best_idx]
    return g_best, p_best_fitness[g_best_idx]

# ========================================
# 2. ANT COLONY OPTIMIZATION (ACO)
# ========================================
def run_aco(dist_matrix, cities, start_nodes=[0, 1, 2], alpha=1.0, beta=2.0, psi=0.5, Q=100.0):
    n = len(cities)
    tau = np.ones((n, n))
    eta = np.zeros((n, n))
    for i in range(n):
        for j in range(n):
            if i != j: eta[i, j] = 1.0 / dist_matrix[i, j]

    tours, lengths = [], []
    for s in start_nodes:
        tour, visited = [s], {s}
        while len(visited) < n:
            curr = tour[-1]
            probs = np.zeros(n)
            denom = 0.0
            allowed = [j for j in range(n) if j not in visited]
            for j in allowed:
                val = (tau[curr, j] ** alpha) * (eta[curr, j] ** beta)
                probs[j] = val
                denom += val
            if denom > 0: probs /= denom
            next_node = np.argmax(probs)
```

```
            tour.append(next_node)
            visited.add(next_node)
        tour.append(s)
        length = sum(dist_matrix[tour[k], tour[k+1]] for k in range(len(tour)-1))
        tours.append(tour)
        lengths.append(length)

    tau_new = (1 - psi) * tau
    delta_tau = np.zeros((n, n))
    for tour, length in zip(tours, lengths):
        deposit = Q / length
        for k in range(len(tour) - 1):
            u, v = tour[k], tour[k+1]
            delta_tau[u, v] += deposit
            delta_tau[v, u] += deposit
    tau_final = tau_new + delta_tau
    return tours, lengths, tau_final

# ========================================
# 3. SIMULATED ANNEALING (SA)
# ========================================
def run_simulated_annealing():
    def obj(x): return x**2 - 6 * x - 13
    x_curr, T, alpha, r_nums = 1, 10.0, 0.5, [0.30, 0.80, 0.20, 0.90]
    x_best, f_best = x_curr, obj(x_curr)
    r_idx = 0
    for i in range(1, 5):
        f_curr = obj(x_curr)
        x_prime = x_curr + 1
        f_prime = obj(x_prime)
        df = f_prime - f_curr
        if df <= 0:
            x_curr = x_prime
        else:
            P = np.exp(-df / T)
            r = r_nums[r_idx]; r_idx += 1
            if r <= P: x_curr = x_prime
        if obj(x_curr) < f_best:
            x_best, f_best = x_curr, obj(x_curr)
        T *= alpha
    return x_best, f_best

# ========================================
# 4. REGULARIZATION (OLS, RIDGE, LASSO, ELASTICNET)
# ========================================
def evaluate_regularization():
    X, y = make_regression(n_samples=500, n_features=50, n_informative=10, noise=0.1,
random_state=42)
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    scaler = StandardScaler()
    X_train_s = scaler.fit_transform(X_train)
    X_test_s = scaler.transform(X_test)

    models = {
        "Linear Regression (No Reg)": LinearRegression(),
```

```
        "Ridge (L2)": Ridge(alpha=1.0),
        "Lasso (L1)": Lasso(alpha=0.1),
        "Elastic Net (L1 + L2)": ElasticNet(alpha=0.1, l1_ratio=0.5)
    }
    results = {}
    for name, model in models.items():
        model.fit(X_train_s, y_train)
        pred = model.predict(X_test_s)
        results[name] = {
            "MSE": mean_squared_error(y_test, pred),
            "R2": r2_score(y_test, pred),
            "coef": model.coef_
        }
    return results
```

4. Use Cases and Results Analysis

4.1 Industrial and Academic Use Cases

- **1. Autonomous Robot Path Routing & Micro-Grid Logistics (ACO):** Ant Colony Optimization constructs dynamic minimum-distance trajectories across non-euclidean cost graphs.
- **2. Hyperparameter Global Optimization (PSO & SA):** Swarm and annealing algorithms optimize hyperparameter manifolds containing multimodal, non-differentiable barriers.
- **3. High-Dimensional Biomarkers & Econometric Feature Selection (L1/ElasticNet):** Regularized models eliminate irrelevant features and prevent multicollinearity in dataset regimes where features exceed samples (P > N).

4.2 Experimental Benchmark Results & Quantitative Logs

Table 1: Swarm Optimization (PSO 1D Benchmark Execution Log)

Objective: Maximize f(x) = -x² + 5x + 20 under c1=1, c2=1, u1=0.213, u2=0.876, w=0

Step / Iteration	Global Best (g_best)	Peak Fitness f(g_best)	Swarm Position Range (x)	Key Trajectory Behavior
Initial Setup (t=0)	2.8000	26.1600	[-9.6, 10.0]	Wide swarm dispersion across domain
Iteration 1	2.4900	26.2499	[1.2624, 3.6928]	Rapid velocity alignment toward g_best
Iteration 2	2.4946	26.2500	[2.3378, 2.6391]	Contraction around theoretical peak

				(x*=2.5)
Iteration 3	2.4988	26.2500	[2.4752, 2.5125]	High-precision convergence to global optimum

Table 2: Simulated Annealing (SA Local Search Convergence Log)

Objective: Minimize f(x) = x² - 6x - 13 starting at x0=1, T0=10.0, α=0.5

Iter	Temp	x_curr	x'	f(x_curr)	f(x')	Δf	P(accept)	Rand r	Decision	Best x
1	10.00	1	2	-18	-21	-3	1.0000	N/A	Accept	2
2	5.00	2	3	-21	-22	-1	1.0000	N/A	Accept	3
3	2.50	3	4	-22	-21	+1	0.6703	0.30	Accept	3
4	1.25	4	5	-21	-18	+3	0.0907	0.80	Reject	3

Optimal Solution Identified: x* = 3, f(x*) = -22

Table 3: Ant Colony Optimization (ACO Tour Routing & Pheromone Matrix)

Nodes: A, B, C, D, E | Parameters: α=1.0, β=2.0, ψ=0.5, Q=100.0, τ0=1.0

Tour Construction Output:

• Ant 1 (Start A): A → B → E → C → D → A | Tour Length = 28.0

• Ant 2 (Start B): B → A → E → C → D → B | Tour Length = 26.0 (Shortest Tour)

• Ant 3 (Start C): C → E → B → A → D → C | Tour Length = 28.0

Final Updated Pheromone Matrix (τ_final):

Node	A	B	C	D	E
A	0.5000	11.4890	0.5000	7.6429	4.3462
B	11.4890	0.5000	0.5000	4.3462	7.6429
C	0.5000	0.5000	0.5000	11.4890	11.4890
D	7.6429	4.3462	11.4890	0.5000	0.5000

E	4.3462	7.6429	11.4890	0.5000	0.5000

Table 4: Regularized Regression Benchmark Performance Metrics

Dataset: 500 samples, 50 features, 10 informative features, Noise = 0.1

Regression Model	Regularization Penalty	Mean Squared Error (MSE)	R^2 Score	Sparsity / Shrinkage Profile
Linear Regression (OLS)	None (0)	0.0126	1.0000	Unconstrained coefficients; vulnerable to noise.
Lasso (L1)	α=0.1 (L1)	0.1390	1.0000	Enforces zero coefficients; exact feature selection.
Ridge (L2)	α=1.0 (L2)	0.5244	1.0000	Smooth coefficient decay; preserves all features.
Elastic Net (L1+L2)	α=0.1, ρ=0.5	172.9885	0.9969	Balances group selection and weight decay.

4.3 Result Summary & Key Findings

• **Optimization Velocity & Accuracy:** Swarm optimization (PSO) converges rapidly to global optima (reaching x=2.4988 vs. true peak x*=2.5 in 3 iterations) due to shared social velocity memory (Gbest).

• **Stochastic Local Minima Escape:** Simulated Annealing successfully avoids sub-optimal traps by accepting uphill worst solutions at high temperatures (e.g., accepting Δf = +1 at T=2.50 when r=0.30 ≤ P=0.6703).

• **Stigmergic Routing Convergence:** In ACO, shorter tours accumulate higher pheromone densities (τ_{A-B} = 11.4890 on the 26.0 length tour), reinforcing optimal paths for subsequent generations.

• **Feature Sparsity vs. Shrinkage:** Lasso (L1) regularization zeroes out non-informative coefficients, yielding an effective sparse model, whereas Ridge (L2) scales down weights uniformly without driving them to absolute zero.

5. Failure Scenarios and Pathological Cases

1. PSO Premature Swarm Collapse: When velocity bounds (Vmax) and inertia weights (w) are set to zero without sufficient exploration coefficients (c1, c2), particles lock into localized clusters and fail to traverse complex multimodal functions.

2. SA Quenching & Freezing: If the cooling rate α is too small (e.g., $\alpha < 0.1$), temperature drops rapidly, causing the algorithm to degenerate into standard greedy local search.

3. ACO Stigmergic Stagnation: High pheromone persistence ($\psi \to 0$) causes early ants to deposit heavy trail concentrations, leading the entire colony to converge prematurely to sub-optimal tours.

4. Lasso Grouped Feature Blindness: In datasets with highly correlated feature groups, Lasso randomly selects a single representative feature and zeroes out the rest, which can degrade stability relative to ElasticNet.

6. Conclusion

For high-dimensional continuous landscapes, Particle Swarm Optimization provides superior convergence efficiency. Simulated Annealing excels in noisy, non-convex local search spaces, while Ant Colony Optimization offers a robust framework for combinatorial routing. In statistical regression, Lasso (L1) is optimal for sparse feature extraction, whereas ElasticNet provides the best balance when features exhibit strong correlation structures.